

Immunomodulatory effect of *Juglans sinensis*, *Psoralea corylifolia*, Cheong-a-hwan extract and cyclosporine A on Th1(IFN- γ)/Th2(IL-4) cytokine balance, eosinophil accumulation in a murine model of asthma

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Abstract

The immune balance controlled by T helper 1 (Th1) and T helper 2 (Th2) is crucial for immunoregulation and its imbalance causes various immune diseases including allergic disorders and asthma. It has been recently proposed that asthma may result from an imbalance between T helper 1 (Th1) and T helper type 2 (Th2) cells. *Juglans sinensis* Dode (family, Juglandaceae), *Psoralea corylifolia* Linn. (family, Fabaceae), and Cheong-a-hwan (herbal prescription composed of *J. sinensis*, *P. corylifolia*), which has been used widely in Korean traditional medicine, has been shown to exhibit various biological activities but its immunoregulatory activities have not been well studied. We investigated these activities of *J. sinensis*, *P. corylifolia* and Cheong-a-hwan on Th1/Th2 cytokine production using an ovalbumin-induced asthma animal model.

Our results have shown that *J. sinensis*, *P. corylifolia* and Cheong-a-hwan and CsA have profound inhibitory effects on the accumulation of eosinophils into airways and blood. Also, they upregulated the production of OVA-specific Th1 cytokine (IFN- γ) and downregulated OVA-specific Th2 cytokine (IL-4) in culture supernatant of spleen cells. These results indicate that *J. sinensis*, *P. corylifolia* and Cheong-a-hwan extracts may be a potential novel therapeutic agent for asthma by modulating the relationship between Th1/Th2 cytokine balance.

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1. Introduction

Bronchial asthma results from excess production of T helper 2 (Th2) cytokines (IL-4, IL-5, IL-13) relative to T helper 1 (Th1) cytokine (IFN- γ) (Factor, 2003). Th1 cells are characterized by the prevalent production of IL-2, IFN- γ , and TNF- β , without IL-4, IL-5, IL-9, and IL-13 production. By contrast, Th2 cells are characterized by the prevalent production of IL-4, IL-5, IL-9, and IL-13 in the absence of IFN- γ . Eosinophils have a crucial role in the pathogenesis of allergic diseases. Clinical and experimental studies have established eosinophilia as a marked sign of allergic disorders including asthma (Rothenberg, 1998). Th2 cells and their secreted products initiate a cascade of events that begins with the recruitment and activation of eosinophils, macrophages, and

other inflammatory cells. The recruited eosinophils produce IL-3, IL-5, IL-6, IL-13, granulocyte-macrophage colony stimulating factor (GM-CSF), tumour necrosis factor (TNF), major basic protein, and eosinophil cationic protein (Epstein, 2006).

IL-4 can directly induce airway hyperresponsiveness, airway and blood eosinophilia in asthmatic patients (Shi et al., 1998). The role of Th1 cytokine IFN- γ in asthma is still a matter of debate: in an earlier study Krug and coworkers described an increased frequency of IFN- γ + T cells in bronchoalveolar lavage fluid from asthmatic compared with control subjects (Krug et al., 1996), and other investigators have shown an inhibitory effect of IFN- γ on pulmonary allergic responses. IFN- γ downregulated the secretion of certain Th2 cytokines. The local administration of aerosolized recombinant IFN- γ in mice prevented antigen-induced eosinophil infiltration into the trachea and normalized airway function (Nakajima, Iwamoto, & Yoshida, 1993).

In our preliminary study, several medicinal herbs have shown to promote immunity in different ways. *Juglans sinensis*

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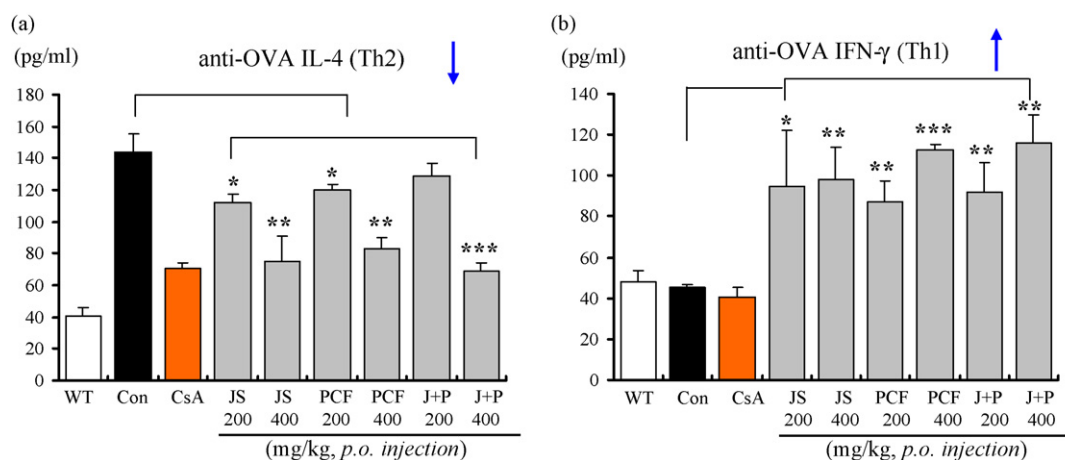


Fig. 1. Immunomodulatory effects of JS, PCF, CAH on OVA-specific Th1/Th2 cytokines production in spleen cells. After 48 h of culture, supernatant from the splenocytes of JS, PCF, CAH fed mice were analyzed the levels of IL-4 and IFN- γ in each samples (described in Section 3). WT: normal C57BL/6 mice; Con: ovalbumin inhalation (control group); CsA: OVA + CsA (10 mg/kg); JS: OVA + JS (200, 400 mg/kg); PCF: OVA + PCF (200, 400 mg/kg); CAH (JS + PCF): OVA + CAH (200, 400 mg/kg). The results are expressed the mean $\pm$ S.E. ($N = 5$). Statistically significant value compared with control group data by t -test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Dode (family, Juglandaceae), *Psoralea corylifolia* Linn. (family, Fabaceae), and Cheong-a-hwan (Cheong-a-hwan; herbal prescription composed of *J. sinensis*, *P. corylifolia* by a ratio of 50:50 based on Korean traditional medicinal formulae) are medicinal herbs and prescription that have been used for treating airway inflammatory diseases (including asthma) in Korea.

They have shown to augment specific cellular and humoral immune response. *P. corylifolia* Linn. has been used traditionally as medicine in China and recommended for the treatment of stomachic, deobstruent, anthelmintic, diuretic, vitiligo and also certain skin diseases, e.g., leucoderma, psoriasis and leprosy (Kotiyal & Sharma, 1992; Zhu, 1998). *P. corylifolia* seed extract has been found to stimulate the immune system in mice (Latha, Evans, Panikkar, & Jayavardhanan, 2000). *J. sinensis* exerts the beneficial effect against mercury chloride-induced acute renal failure and its effect may be due to antioxidant action (Ahn, Song, Kim, & Kim, 2002).

Few reports, however, have addressed the immunoregulatory effects of Cheong-a-hwan, *P. corylifolia* Linn. and *J. sinensis* on Th1/Th2 cytokine balance. Therefore, the aim of this study is to evaluate the immunomodulatory effect of Cheong-a-hwan, *P. corylifolia* Linn. and *J. sinensis* and to understand the effects of eosinophil migration.

2. Results and discussion

2.1. Immunomodulation of OVA-specific Th1/Th2 cytokines production in spleen cells

Spleen cells were isolated from experimental murine model of asthma (in Section 3) and stimulated in the absence JS, PCF, CAH and CsA.

To study whether JS, PCF, CAH were related to Th1/Th2 cytokine balance, after 48 h of culture, supernatant from the splenocytes of JS, PCF, CAH fed mice were analyzed the levels of IL-4 and IFN- γ in each samples.

As shown in Fig. 1, IL-4 productions in spleen were suppressed by JS, PCF, CAH dose dependently. On the contrary, JS, PCF, CAH dose dependently enhanced the secretion of IFN- γ levels.

2.2. Inhibitory effects of JS, PCF, CAH and CsA on eosinophil accumulation in BALF, blood

JS, PCF, CAH and CsA also attenuated the decrease of eosinophils in BALF induced by repeated i.n. administration of allergen (OVA inhalation 500 μ g/ml; on two or three consecutive days). The number of eosinophils was significantly different among these three groups. In OVA-exposed group, the number of eosinophils was significantly increased, but in JS, PCF, CAH treated group, the number of it were significantly decreased in comparison with the OVA-exposed group (Figs. 2 and 3).

JS, PCF and CAH are well known herb medicines and prescription used in oriental medicine for treatment anti-inflammatory and many allergic diseases. They were prepared by the routine method of water decoction, which is in accordance with the method used in clinical application.

Airway inflammation is associated with the infiltration of eosinophils, lymphocytes into the airways and increased numbers of these cells have been found in the bronchoalveolar lavage fluid (BALF) and bronchial biopsies from asthmatics (Kay, 1991). It has been suggested that eosinophils contribute to several of the clinical features of allergic asthma, including tissue damage and airway hyperresponsiveness (Bousquet et al., 1990). Th2 cytokines cause recruitment of eosinophils to the airway. Also, Th2 cytokine IL-4 plays a role in the pathogenesis of allergy and asthma (Mosmann & Sad, 1996) and Interferons are Th1-type cytokines that have a role in inhibiting IgE production and Th2-cell proliferation and function (Busse & Rosenwasser, 2003).

Biological balance appears to be needed between IFN- γ and IL-4 producing Th1 and Th2 subsets so that health can be maintained and autoimmune disease can be prevented.

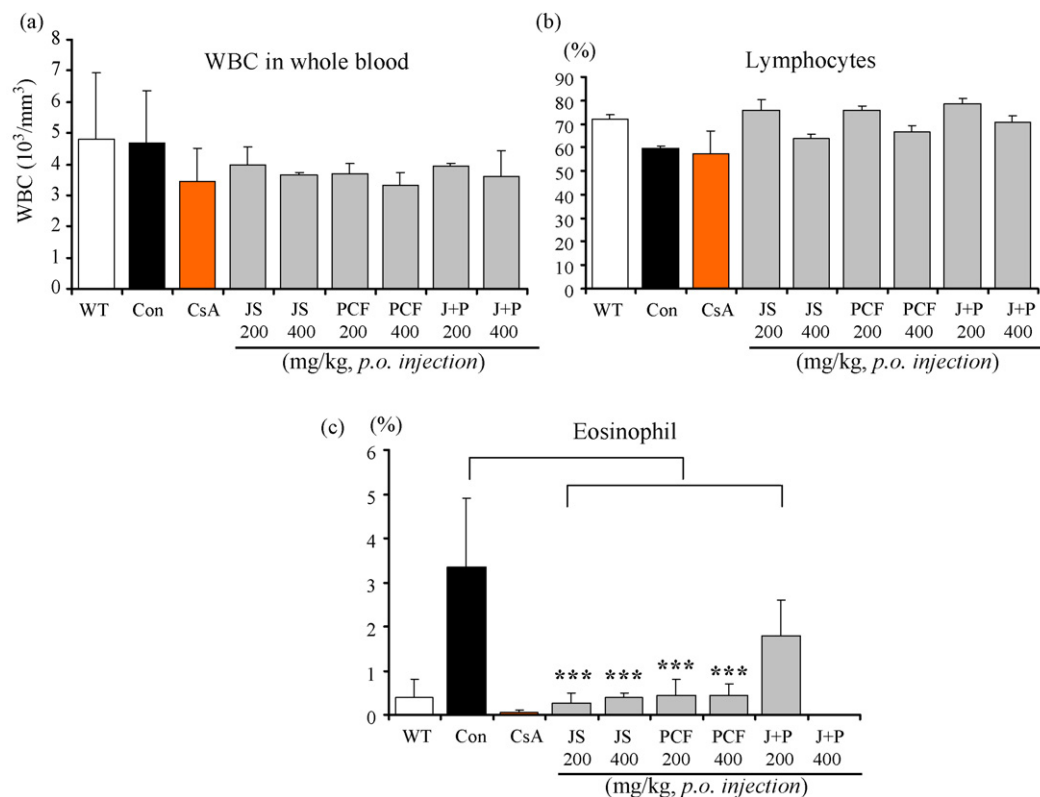


Fig. 2. Effects of JS, PCF, CAH on total WBC, lymphocytes and eosinophil cell levels in blood. Mice were processed as described in Section 3, and whole blood was harvested 24 h after the last OVA challenge. Total inflammatory cell numbers in blood were counted, and cell classification was performed on a minimum of 200 cells to classify eosinophils and lymphocytes. Data are expressed as mean $\pm$ S.E.M. ($n = 5$). WT: normal C57BL/6 mice; Con: ovalbumin inhalation (control group); CsA: OVA + CsA (10 mg/kg); JS: OVA + JS (200, 400 mg/kg); PCF: OVA + PCF (200, 400 mg/kg); CAH (JS + PCF): OVA + CAH (200, 400 mg/kg). The results are expressed the mean $\pm$ S.E. ($N = 5$). Statistically significant value compared with control group data by t -test (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Cyclosporine A (CsA) has been shown to inhibit single allergen-induced allergic inflammation such as eosinophilic and lymphocytic infiltration and mRNA expression for interleukin (IL)-4 and IL-5.

In our study, the JS, PCF and CAH could modulate the immune responses of Th1/Th2 cytokine production in spleen cells in OVA-induced murine model of asthma. One of the therapeutic objectives in asthma is to reduce the regional inflammatory response through the reduction of antigen-induced inflammatory cells and inflammatory cytokines production. Moreover, diagnosis of Th1/Th2 balance in asthma is essential for the application of immune balance regulating drugs. The diagnosis of Th1/Th2 balance is also useful for early detection of recurrence or cure of immune diseases (including asthma).

Th1/Th2 cytokine imbalance contribute to the pathogenesis of asthma, understanding these mechanisms will provide novel targets for therapeutic activities.

Therefore, we investigated the effects of JS, PCF and CAH on Th1/Th2 balance especially between IFN- γ and IL-4 production in spleen of murine asthma model.

Our data showed that the immunomodulatory mechanisms of JS, PCF and CAH on Th1/Th2 cytokine balance may have involved: Reduced production of IL-4 and enhancement of IFN- γ . This was accompanied by changes of eosinophils counts in BALF and blood (Figs. 2 and 3).

The use of drugs (including antihistamines) to modulate the Th1/Th2 cytokine balance in the prevention of asthma remains attractive. Aside from the use of drugs and cytokines to alter the Th1/Th2 balance, several other strategies are under current investigation. One such strategy is the invention of novel anti-allergic herb medicine and prescription to modulate the Th1/Th2 balance.

The potential therapeutic activity of JS, PCF and CAH on asthma in oriental medicine may be partly related to modulate the relationship between Th1/Th2 cytokine balance. However, Th1/Th2 balance is not only controlled by Th1 cells and Th2 cells, but controlled by various regulatory factors including regulatory T cells, sexual factor, chemokines, transcription factors (including NF- κ B, AP-1), signal transduction pathway (STATs), etc.

The Th1/Th2 balance could be influenced by immunomodulatory drugs (including herbs, prescription and its main components) but this way of therapy needs further evaluation.

3. Materials and methods

3.1. Plant material and preparation of extracts

The sample of *Juglans sinensis* Dode (family, Juglandaceae), *Psoralea corylifolia* Linn. (family, Fabaceae), and Cheong-a-hwan were purchased from Saerom company

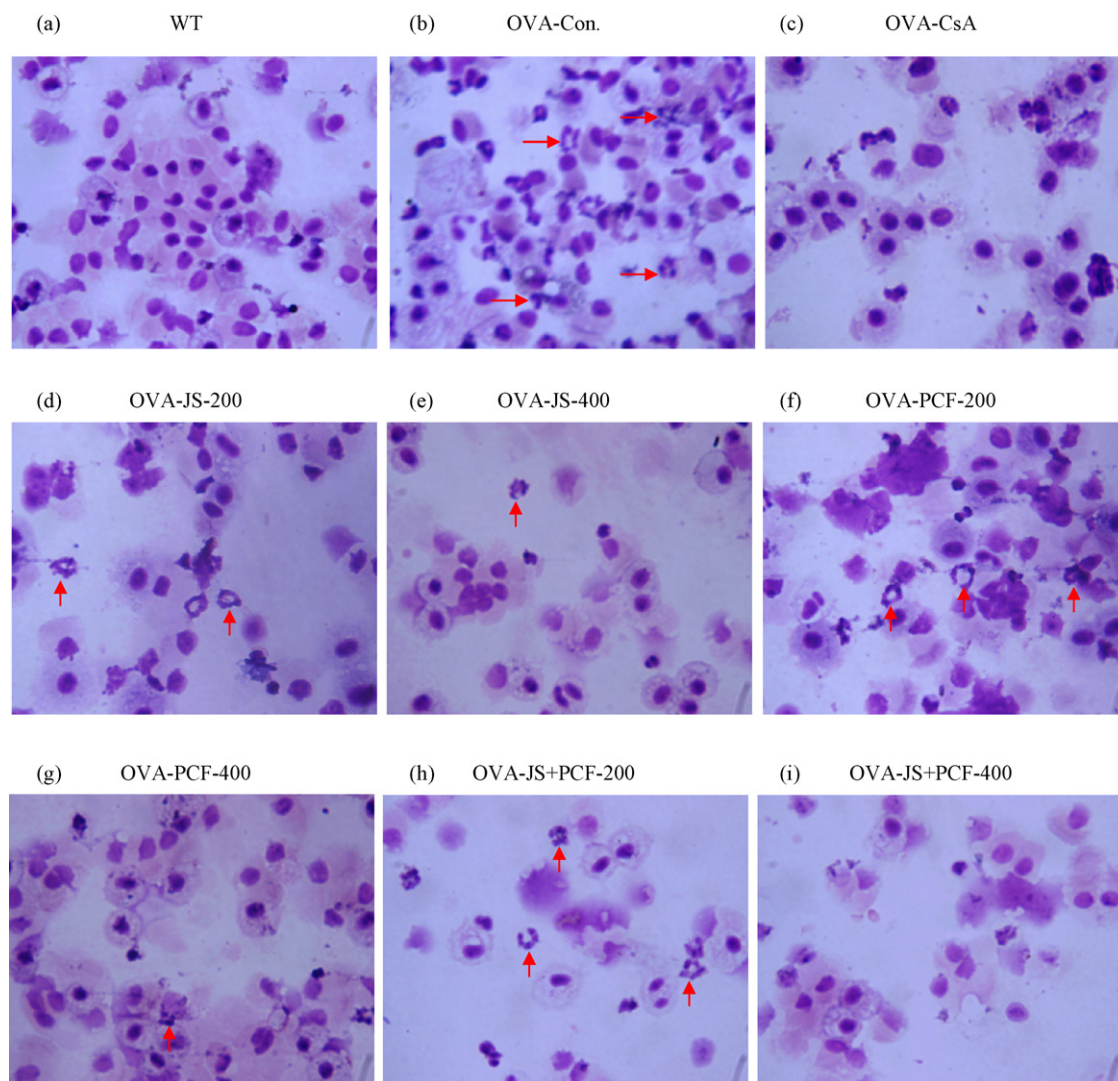


Fig. 3. Photomicrograph (original magnification, $\times 400$) of BALF cytopins from a sensitized mouse exposed repeatedly to allergen (OVA; 500 $\mu\text{g}/\text{ml}$, i.n. for 30 min/day on 3 days/week for 12 weeks at a flow rate of 250 L/min, 2.5% ovalbumin in normal saline). WT: normal C57BL/6 mice; Con: ovalbumin inhalation (control group); CsA: OVA + CsA (10 mg/kg); JS: OVA + JS (200, 400 mg/kg); PCF: OVA + PCF (200, 400 mg/kg); CAH (JS + PCF): OVA + CAH (200, 400 mg/kg).

(Ansung, Korea) in February, 2006. The plant was identified by Professor Young-Cheol Lee, College of Oriental Medicine, Sangji University in Wonju, Korea, and a voucher specimens (*J. sinensis*; JS, *P. corylifolia*; PCF, and Cheong-a-hwan; CAH) are deposited in our laboratory (Department of Herborology, College of Oriental Medicine, Sangji University Wonju 220–702, Republic of Korea). Plant material (250 g) was extracted three times with distilled water. Then, the extract was filtered and evaporated on a rotatory evaporator (Rotary evaporator, BUCHI B-480, Switzerland) and finally dried by a freeze drier (Freeze dryer, EYELA FDU-540, Japan) to yield the extract JS (25 g), PCF (20 g) and CAH (18 g).

The yield (w/w) of the extract was about 10% (JS), 8% (PCF) and 7.2% (CAH).

3.2. Animals

Seven to eight-week-old male C57BL/6 mice were obtained at Daehan Biolink Co. Ltd. (Eumsung, Republic of Korea). All

animal procedures were conducted in accordance with the guidelines of the Institutional Animal Care and Use Committee, Korea Research Institute of Bioscience and Biotechnology (Daejeon, Republic of Korea).

3.3. OVA sensitization and inhalation

OVA (500 $\mu\text{g}/\text{ml}$) in PBS was mixed with equal volumes of 10% (w/v) aluminum potassium sulfate (alum; Sigma) in distilled water. Then incubated for 60 min at RT after adjustment to pH 6.5 using 10N NaOH, and centrifuged at $750 \times g$ for 5 min. OVA/alum pellet was resuspended to the original volume in distilled water. All mice were immunized on two different days (21 days and 7 days before inhalational exposure) by i.p. injections of 0.2 ml alum-precipitated Ag containing 100 μg of OVA (Sigma–Aldrich Korea, Korea) bound to 4 mg of aluminum hydroxide (Sigma–Aldrich Korea, Korea) in PBS. Seven days after the second sensitization, mice were exposed to aerosolized ovalbumin for 30 min/day on 3

days/week for 12 weeks (at a flow rate of 250 L/min, 2.5% ovalbumin in normal saline) and intratracheally injected 250 µg of OVA (on day 8) on the back of the tongue. JS (200, 400 mg/kg), PCF (200, 400 mg/kg), CAH (200, 400 mg/kg) and CsA (10 mg/kg) in solution form were orally administered three times a week for last 8 weeks. One day after the last of the OVA exposures, samples (BALF, spleen and blood) were collected. Total and differential WBC counts in blood were determined within 24 h after sample collection by means of automated hematology analyzer.

3.4. BALF

To evaluate airway inflammation, we experimented the accumulation of eosinophils in BALF. Mice were sacrificed with an intraperitoneal injection of sodium pentobarbitone (100 mg/kg). Immediately after sacrifice, cells in the lungs were recovered by flushing 1 ml of BAL fluid (1 mM EDTA, 10% FBS, PBS) into the lungs via the trachea. Total cell counts were determined and 100 µl of fluid were cytospun onto glass slides using a Cytospin centrifuge (Cellspin, Hanil, Korea) (400 g for 4 min). Differential cell counts were performed after staining with Diff-Quik Stain Set (Baxter Healthcare Corp., Miami, Florida, USA).

3.5. Enzyme-linked immunosorbent assay (ELISA)

OVA-specific Interleukin-4 (IL-4) and IFN-γ production from spleen cells of the indicated mice ($n = 5$) was measured by ELISA according to the manufacturer's instruction on a monoclonal antibody-based mouse interleukin ELISA kit (R&D system). Briefly, after washing three times, the spleen cells were suspended in RPMI 1640 medium supplemented with 2 mM L-glutamine, 5% fetal bovine serum. The cells were then cultured for 48 h at 1×10^5 cells/well in 96-well culture plates (Corning Inc., Cambridge, Mass) with or without 1 µg/ml of OVA in a humidified atmosphere of 5% CO₂ in air at 37 °C. The culture supernatants were collected and were assayed for IFN-γ, IL-4 antibody to OVA by ELISA.

All data represent the standard deviation of at least three different determinants and were compared using Student's paired *t*-test.

3.6. Statistical analysis

For statistical analysis of data, *P*-values were analyzed using a paired Student's *t*-test software program (Startview 5.1; Abacus Concepts, Berkeley, CA). Results were considered statistically significant when *p*-values were (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

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References

- Ahn, C. B., Song, C. H., Kim, W. H., & Kim, Y. K. (2002). Effects of *Juglans sinensis* Dode extract and antioxidant on mercury chloride-induced acute renal failure in rabbits. *J. Ethnopharmacol.*, 82, 45–49.
- Bousquet, J., Chanez, P., Lacoste, J. Y., Barnéon, G., Ghavanian, N., Enander, I., et al. (1990). Eosinophilic inflammation in asthma. *N. Engl. J. Med.*, 323, 1033.
- Busse, W. W., & Rosenwasser, L. J. (2003). Mechanisms of asthma. *J. Allergy Clin. Immunol.*, 111, S799–S804.
- Epstein, M. M. (2006). Targeting memory Th2 cells for the treatment of allergic asthma. *Pharmacol. Therapeut.*, 109, 107–136.
- Factor, P. (2003). Gene therapy for asthma. *Mol. Ther.*, 7, 148–152.
- Kay, A. B. (1991). Asthma and inflammation. *J. Allergy Clin. Immunol.*, 87, 893–910.
- Kotiyal, J. P., & Sharma, D. P. (1992). Phytochemical studies of *Psoralea* species. *Bull. Med.-Ethnobotanical Res.*, 13, 209–223.
- Krug, N., Madden, J., Redington, A. E., Lackie, P., Djukanovic, R., Schauer, U., et al. (1996). T-cell cytokine profile evaluated at the single cell level in BAL and blood in allergic asthma. *Am. J. Respir. Cell Mol. Biol.*, 14, 319–326.
- Latha, P. G., Evans, D. A., Panikkar, K. R., & Jayavardhanan, K. K. (2000). Immunomodulatory and antitumour properties of *Psoralea corylifolia* seeds. *Fitoterapia*, 71, 223–231.
- Mosmann, T. R., & Sad, S. (1996). The expanding universe of T-cell subsets: Th1, Th2 and more. *Immunol. Today*, 17, 138–146.
- Nakajima, H., Iwamoto, I., & Yoshida, S. (1993). Aerosolized recombinant interferon-gamma prevents antigen-induced eosinophil recruitment in mouse trachea. *Am. Rev. Respir. Dis.*, 148, 1102–1104.
- Rothenberg, M. E. (1998). Eosinophilia. *N. Engl. J. Med.*, 28, 1592.
- Shi, H. Z., Deng, J. M., Xu, H., Nong, Z. X., Xiao, C. Q., Liu, Z. M., et al. (1998). Effect of inhaled interleukin-4 on airway hyperreactivity in asthmatics. *Am. J. Respir. Crit. Care Med.*, 157, 1818–1821.
- Zhu, Y. P. (1998). *Chinese materia medica: chemistry, pharmacology and applications*. Amsterdam: Harwood Academic Publishers. pp. 609–612.